FED-STD-H28/14A
June 9, 1983
SUPERSEDING
FED-STD-H28/14
August 31, 1978
(See Note)

FEDERAL STANDARD

SCREW-THREAD STANDARDS FOR FEDERAL SERVICES

SECTION 14

BUTTRESS SCREW THREADS–7°/45° FLANK ANGLES

This standard was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

NO DELIVERABLE DATA REQUIRED BY THIS DOCUMENT

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INFORMATION SHEET ON FEDERAL STANDARDS

This Federal Standard is issued in loose-leaf form to permit the insertion or removal of new or revised pages and sections.

All users of Federal Standards should keep them up to date by inserting revised or new pages as issued and removing superseded and canceled pages.

New and revised pages will be issued under Change Notices which will be numbered consecutively and will bear the date of issuance. Change Notices should be retained and filed in front of the standard until such time as they are superseded by a reissue of the entire standard.

FOREWORD

This section was developed to provide Buttress Screw Threads for the Federal Services. It was formerly known as "National Buttress Threads." The present issue is a complete revision of FED-STD-H28/14 dated 31 August 1978.

FED-STD-H28/14A was prepared by the Defense Industrial Supply Center (DLA-IS) and incorporates the American National Standard for Buttress Inch Screw Threads, ANSI B1.9-1973. Significant changes from the previous issue include the following:

(1) The Class 1 (free) thread has been eliminated. This thread had a pitch diameter tolerance 50% larger than the standard grade Class 2 thread tolerance.

(2) Tolerances on major diameter of external threads and minor diameter of internal threads have been increased. They are now the same as the pitch diameter tolerances.

(3) Tolerances have been added for Class 3 (precision grade) for thread lead, flank angles, taper and roundness.

(4) A less expensive, flat root form thread has been added as an option.

(5) Greater detail has been provided for acceptability gaging and gages.

(6) Additional information for British form buttress threads has been included.

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SECTION 14 - BUTTRESS SCREW THREADS - 7°/45° LOAD AND CLEARANCE ANGLES

1. Scope. This section provides the standards for Buttress type screw threads, their gages and acceptability requirements. Buttress threads feature load flanks nearly perpendicular to the thread axis. This minimizes the radial component of a single direction axial force applied to the load flanks.

1.1 Limitations. Threads covered in this section are limited to those with load flank angles of 7° and clearance flank angles of 45°.

1.2 Classification. The following thread forms are included in this section:

American National Standard Form - Engagement height 0.6 x pitch
British Standard Form - Engagement height 0.4 x pitch

2. Referenced documents.

2.1 Government publications. The issue of the following document in effect on the date of invitation for bids or request for proposal forms a part of this standard to the extent specified herein.

Federal standards.

FED-STD-H28/1 *Nomenclature, Definitions and Letter Symbols for Screw Threads*

(Activities outside the Federal Government may obtain copies of Federal specifications and standards as stated in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, including cumulative supplements issued during the year, is sold on subscription by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of Federal specifications and standards required for bidding purposes are available from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA, or from the General Services Administration Supply Distribution Facility Franconia-Building A, Loisdale Road, Franconia, VA 22105.

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2.2 Other publications. The following document forms a part of this standard to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard

ANSI B1.9-1973 Buttress Inch Screw Threads

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017 or the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. Definitions. The terms applicable to this standard are defined in FED-STD-H28/1.

4. General requirements. Not applicable.

5. Detailed requirements.

5.1 American National Standard Form Buttress Threads.

5.1.1 This thread form is used for general purpose applications of buttress threads.

5.1.2 Thread design, gages, acceptability, designation, etc. shall be in accordance with the requirements of ANSI B1.9-1973.

5.2 British Standard Form Buttress Threads.

5.2.1 This thread form is used for special purpose applications of buttress threads where tube wall is too thin or where fatigue requirements are too great for the use of the American National Standard Form buttress threads.

5.2.2 The following requirements apply:

a. The design is in accordance with Appendix C of ANSI B1.9-1973.

b. Thread series is in accordance with Table 1 of ANSI B1.9-1973 except that smallest size is 1 inch diameter. Pitches equivalent to 5, 2 1/2 and 1 1/4 threads per inch are not preferred.

c. Thread tolerances and allowance are in accordance with Sections 6 and 7, respectively, of ANSI B1.9-1973. British medium class is Class 2 and close class is Class 3.

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d. Gaging practice, unless otherwise specified, is in accordance with Section 11 of ANSI B1.9-1973. Gage minor diameter must be increased by 0.4p due to difference in profile.

e. Thread acceptability is in accordance with paragraph 12.1 of ANSI B1.9-1973 except that root radius inspection is required. Radius of curvature shall not be less than the specified radius at any point.

f. Thread designation, in accordance with British practice, includes only nominal size, B.S. buttress thread, threads per inch and class. It is recommended that for greater definition, American practice in accordance with Section 9 of ANSI B1.9-1973 be used except that BUTT and PUSH BUTT shall be replaced by B.S. BUTT and B.S. PUSH BUTT, respectively.

NOTE: Flat root threads, designated FL, are not standard.

5.2.3 Additional information concerning British buttress threads.

a. Pitch diameter equivalents for flank angle deviations do not conform to Appendix A or Table 10 of ANSI B1.9-1973 due to the reduced thread height. British practice gives the pitch diameter equivalents as $\Delta E\alpha_{12} = p[0.0065\alpha_2 + 0.0125\alpha_1]$ where $\delta\alpha_1$ is the variation in the 45° angle and $\delta\alpha_2$ is the variation in the 7° angle. $\delta\alpha_1$ and $\delta\alpha_2$ are in degrees or fractions of a degree.

b. Pitch diameter measurement practices in Appendix B of ANSI B1.9-1973 do not apply in full to the British buttress threads. Recommended minimum (British best) size wires for these threads contact the load flank at the pitch line. The following formulas apply for this condition:

$$w_{min} = 0.43766p$$

Projection, $a = 0.0455p$

Wire Constant, $C_{min} = 0.491011p$

Maximum wire sizes listed in Table 11 of ANSI B1.9-1973 are excessive for the 0.4p thread engagement height. Recommend that best wire sizes listed in Table 11 be used as maximum sizes with projection, a, increased to 0.2094p and C_{best} applicable.

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6. Notes.

6.1 Supersession note. In addition to superseding FED-STD-H28/14, dated 31 August 1978, this document also supersedes part of Appendix A13 of FED-STD-H28, dated 31 March 1978, where it applies to buttress threads.

MILITARY INTERESTS:

Custodians:

Army - AR
Navy - AS
Air Force - 11

Review Activities:

Army - AT, AV, MI
Air Force - 15, 80, 82, 99

User Activity:

Navy - EC

CIVIL AGENCY COORDINATING ACTIVITIES:

Commerce - NBS
DOT - AAF, ACO, FAA, FRA, NHT,
GSA - FSS PCD
HUD - HEP
Justice - FPI
NASA - JFK, LRC, MSF
USDA - AFS

PREPARING ACTIVITY:

DLA - IS

(DoD Project THDS -0040)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
FED-STD-H28/14A	BUTTRESS SCREW THREADS-7° /45° FLANK ANGLES

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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